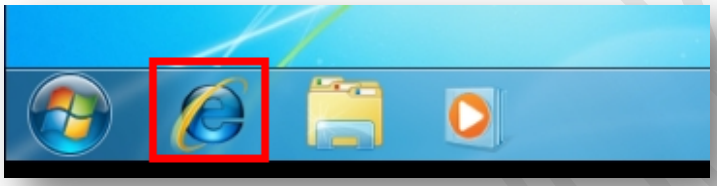
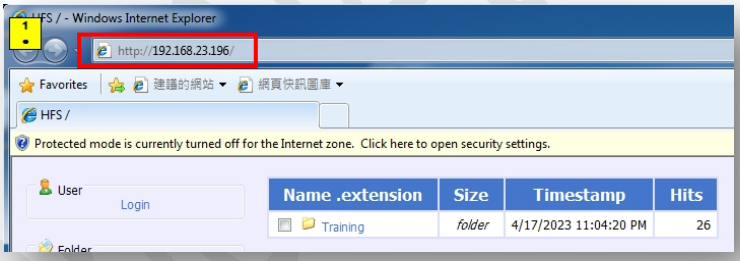
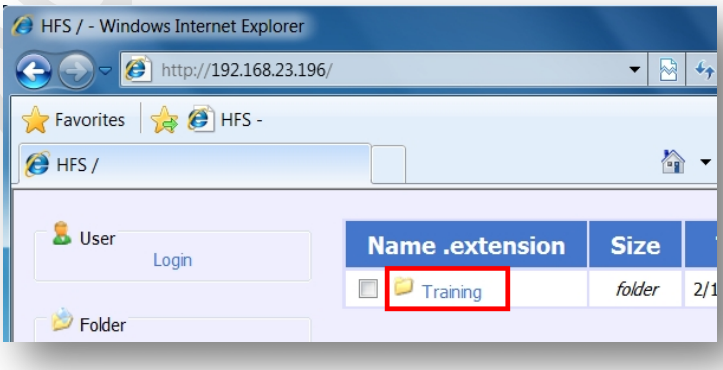
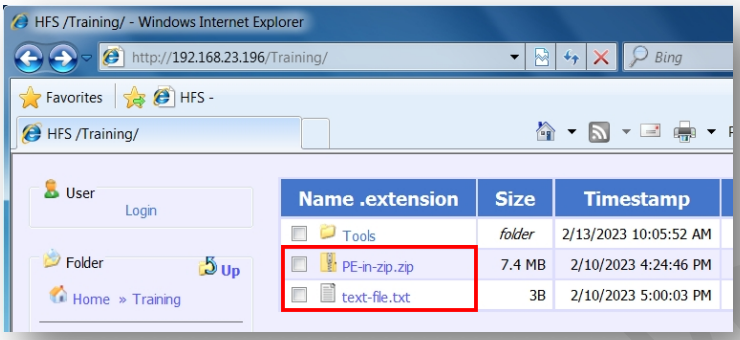
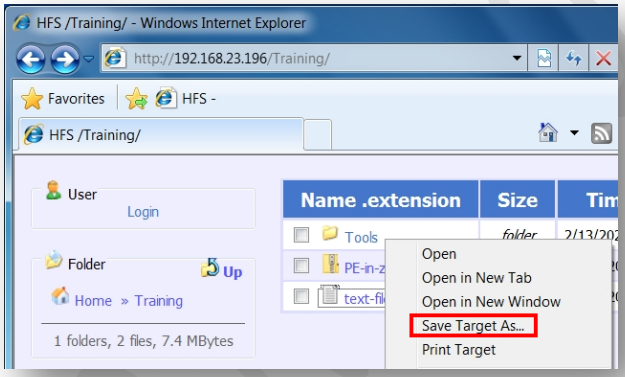
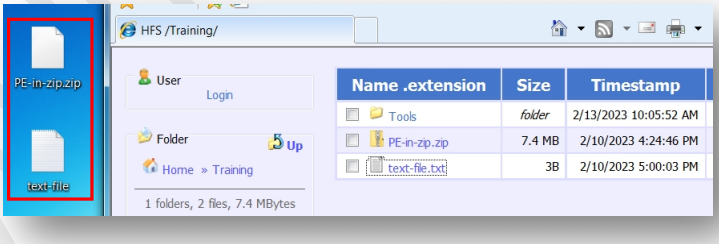
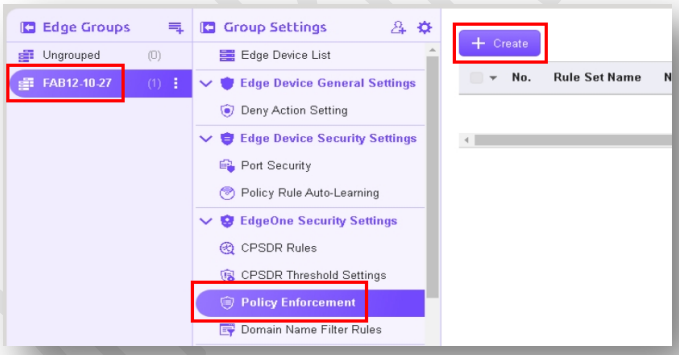
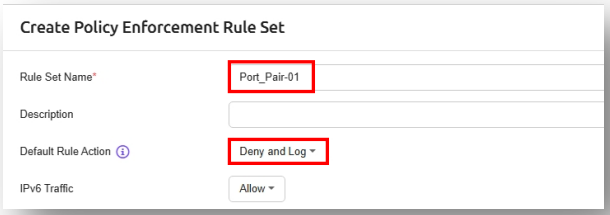
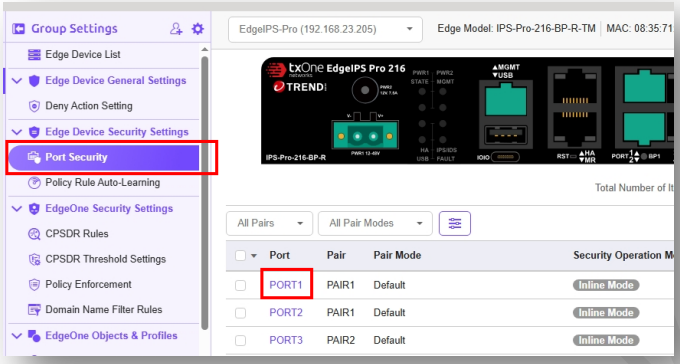
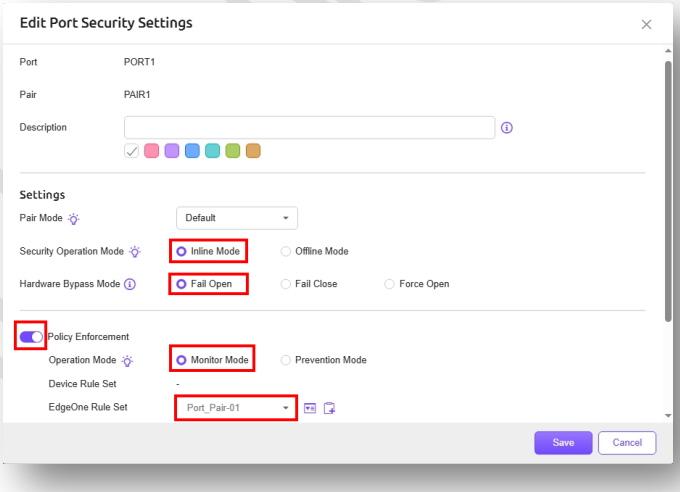
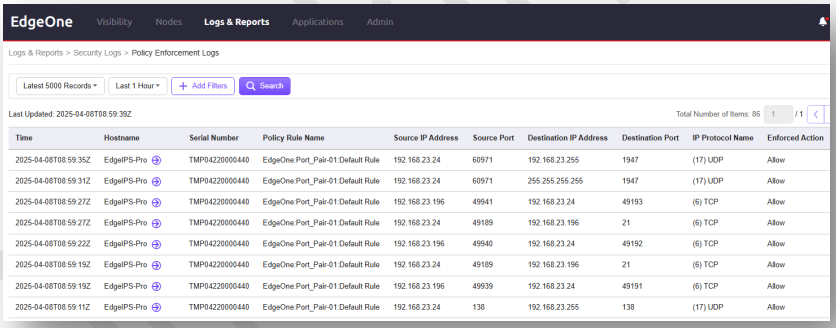
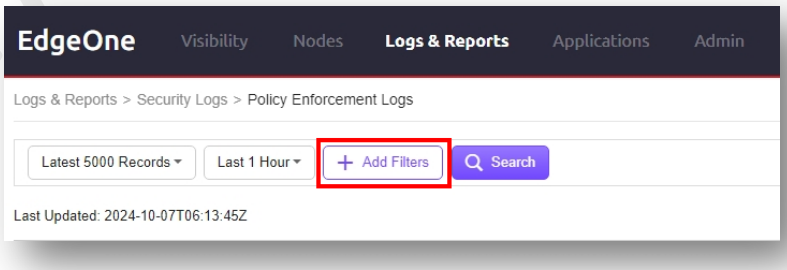
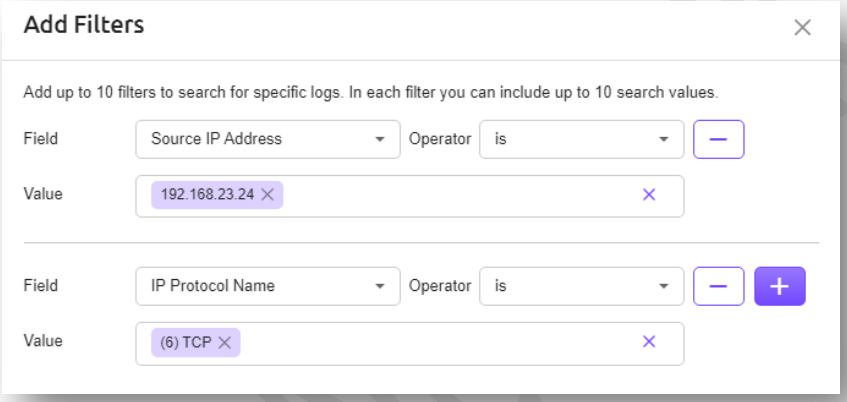
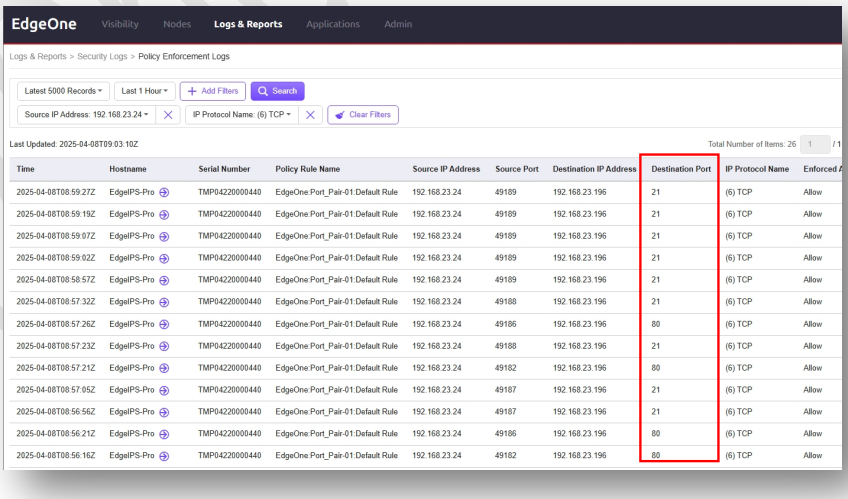
Items	Descriptions
	7. Enter "**quit**" to exit the FTP client 8. Enter "**exit**" to exit the CMD 9. Delete the **PE-in-zip.zip** and **text-file.zip** files from the desktop. **Download files over HTTP.** 1. On the VM **Win7**, which should run on **Host B**, Launch the Internet Explorer (IE) browser from the taskbar.  2. IE will access the HFS web server automatically (**Which runs on Host A**). Alternatively, manually go to: http://192.168.23.196  3. Click on the "**Training**" web folder. 

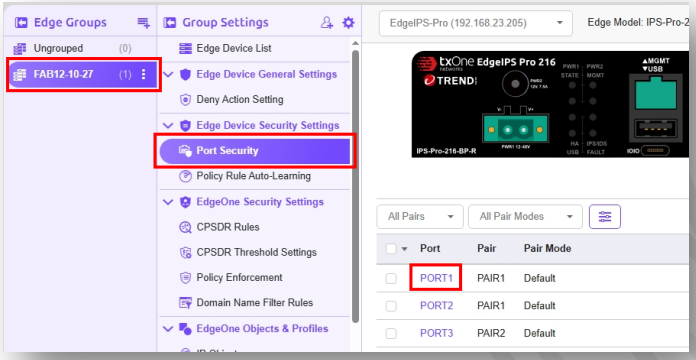
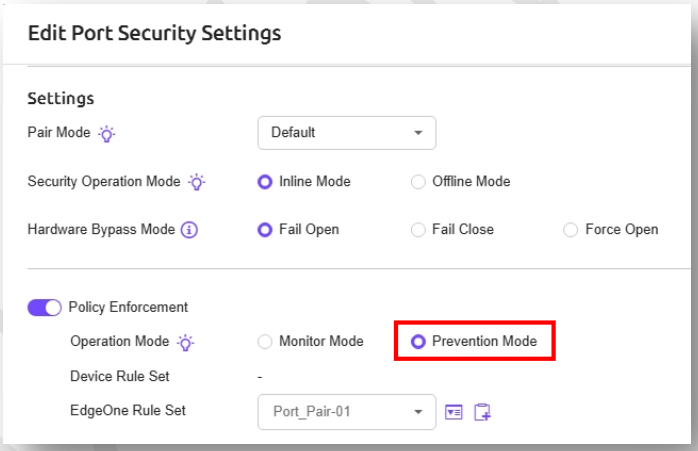
Items	Descriptions
	4. Download **PE-in-zip.zip, text-file.txt** and store the files on desktop.  5. To download the text file, right click it and choose "**Save Target As…**"  6. Check if the files successfully downloaded to the desktop or not.  7. Close the IE browser. 8. Delete the **PE-in-zip.zip** and **text-file.txt** from the desktop. 9. This step was simply to observe the system's behavior without EdgeIPS Pro 216 in the connection.

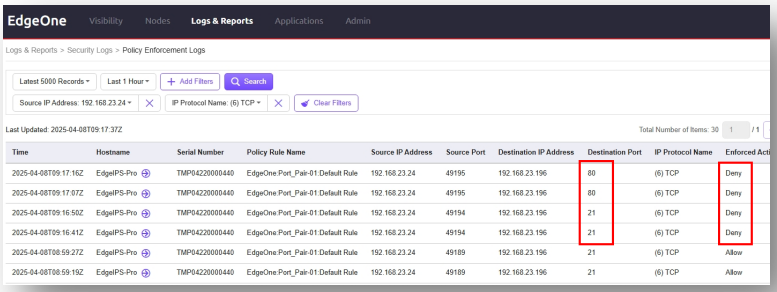
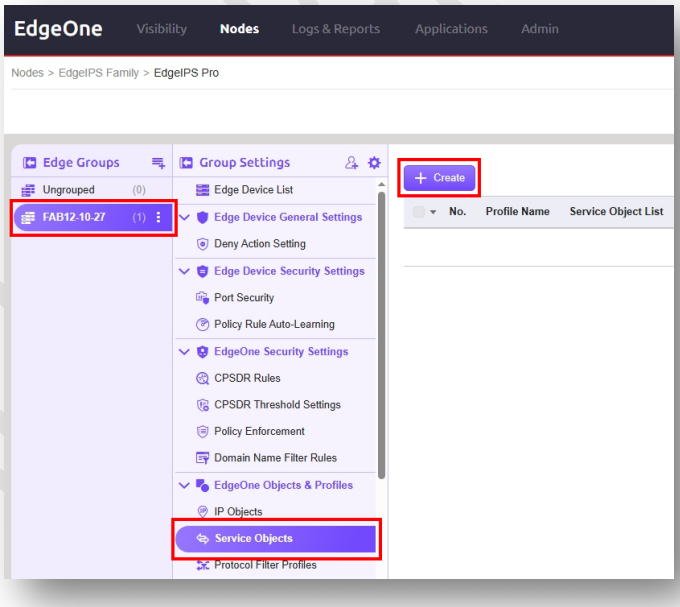
Items	Descriptions
	Monitor Mode versus Prevention Mode **Objective :** 1. Demonstrate FTP and HTTP access are functional under **Monitor** Mode. 2. Demonstrate that, after switching to **Prevention** Mode, access is blocked (Deny All) **Steps:** 1. Log in to the EdgeOne Web UI, go to https://192.168.23.21 or click the "**EdgeOne**" bookmark in Chrome. 2. Go to **Node > EdgeIPS Family > EdgeIPS Pro** 3. Select the Edge Group "**FAB12-10-27**" 4. Choose "**Policy Enforcement**" 5. Click "**+ Create**"  6. Create a Policy Enforcement Rule Set for Port pair1 Name: Port_Pair-01 Default Rule Action: Deny and Log  7. Click "**Save**"

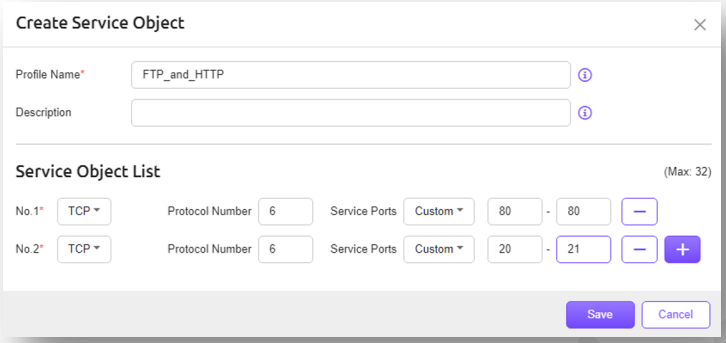
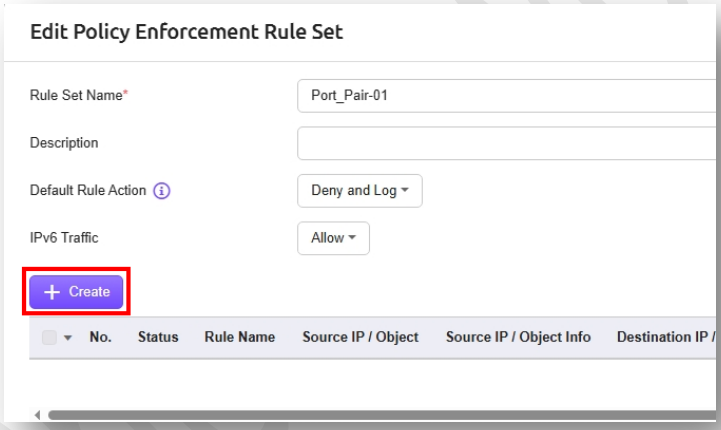
Items	Descriptions
	8. Choose “**Port Security**” and click “**Port1**”  9. Configure the Port Security Settings - Security Operation Mode: **Inline Mode** - Hardware Bypass Mode: **Fail Open** - Policy Enforcement: **Enabled** - Operation Mode: **Monitor Mode** - EdgeOne Rule Set: **Port_Pair-01**  10. Click “**Save**”

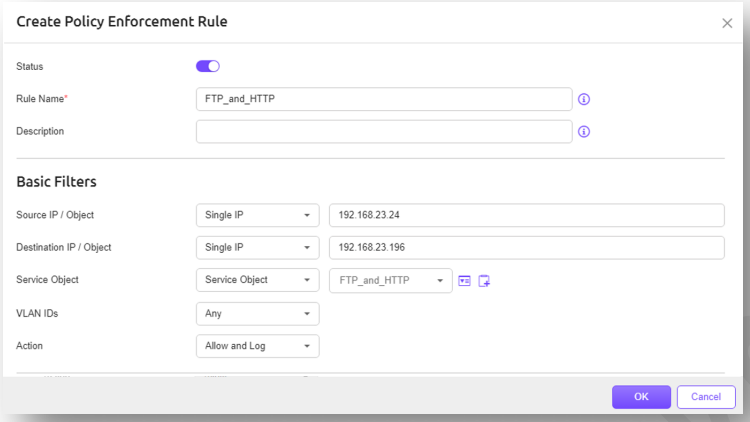
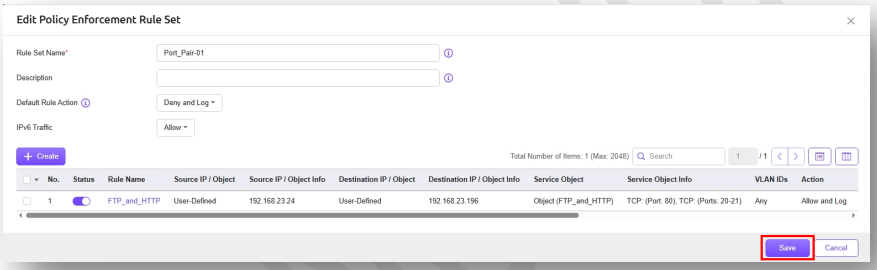
Items	Descriptions
	11. Verify the Port Security configuration 12. Execute the "**Download files over FTP**" SOP. The files should be successfully downloaded by FTP. 13. Execute the "**Download files over http"** SOP. The files should be successfully downloaded over HTTP. 14. Go to **Logs & Reports > Security Logs > Policy Enforcement Logs** to check the logs.  15. You can apply event filters if you see too much "noise". Click on the "**Add Filter(s)**" button. 

Items	Descriptions
	16. Entry 1st filter Field: Source IP Address Operator: **is** Value: 192.168.23.24 **17. Entry 2nd filter** Field: IP Protocol Name Operator: **is** Value: **(6) TCP**  18. With the filters applied, the logs are easier to check. 

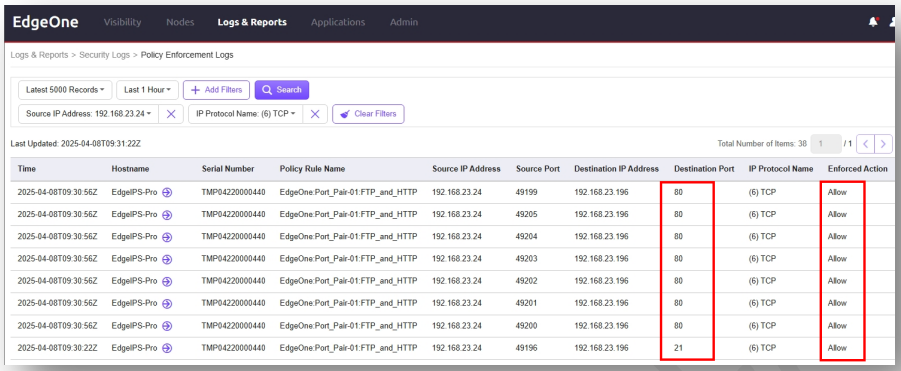
Items	Descriptions
	19. Go to **Node > EdgeIPS Family > EdgeIPS Pro** 20. Select the Edge Group "**FAB12-10-27**" 21. Choose **Port Security** 22. Click **Port 1**  23. Set Operation Mode to "**Prevention Mode**"  24. Remember to click "**Save**" to save the changes. 25. Execute the "**Download files over FTP**" SOP. The file download over FTP should **Fail**. 26. Execute the "**Download files over http**" SOP. The file download over http should **Fail**. 27. Go to **Logs & Reports > Security Logs > Policy Enforcement Logs** to check the logs. **Hint:** You can apply filter rules to filter the logs to find the result easier.

Items	Descriptions
	 **Configure a "Basic filter rule" to allow ftp and http access.** Create a "**Service Object**" for ftp and http 1. Go to **Node > EdgeIPS Family > EdgeIPS Pro** 2. Select the Edge Group "**FAB12-10-27**" 3. Choose the "**Service Object**" 4. Click "**+ Create**"  5. Edit the Profile Name: **FTP_and_HTTP** - No1: Protocol: **TCP**, Service Port: **Custom,80 ~ 80** - No2: Protocol: **TCP**, Service Port: **Custom,20 ~ 21** 6. Click "**Save**"

Items	Descriptions
	 7. Choose the "**Policy Enforcement**" 8. Select the "**Port_Pair-01**" rule set 9. Click "**+Create**"  10. Rule Name: **FTP_and_HTTP** - Source IP / Single IP: **192.168.23.24** - Destination IP / Single IP: **192.168.23.196** - Service Object: Select "**FTP_and_HTTP** - Action: **Allow and Log** 11. Click "**OK**"

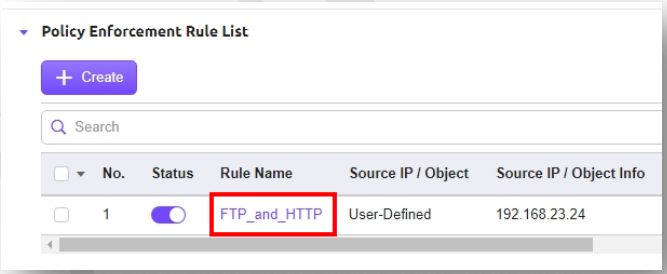
Items	Descriptions
	 12. Click “**Save**” to apply the policy.  13. Execute the “**Download files over FTP**” SOP. The file download over FTP should be **successful**. 14. Execute the “**Download files over http**” SOP. The file download over http should be **successful**. 15. Go to **Logs & Reports > Security Logs > Policy Enforcement Logs** to check the logs. **Tips:** You can add a search filter Rule Name:**FTP_and_HTTP** to make it easier to check the logs.

Items	Descriptions



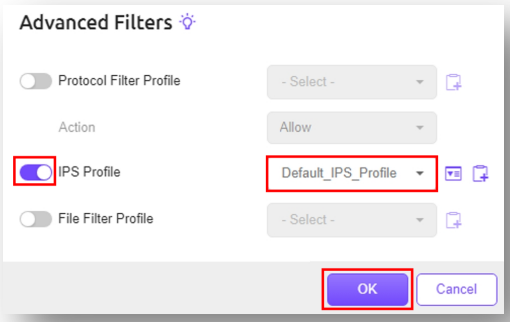
Configure Advanced Filter – IPS Profile to detect suspicious / malicious behavior.

1. Go to **Nodes > EdgeIPS Family > EdgeIPS Pro**
2. Select the Edge Group **: FAB12-10-27**
3. Choose the **Policy Enforcement**
4. Select the **Port_Pair-01** rule set
5. Click on the Rule Name: "**FTP_and_HTTP**"

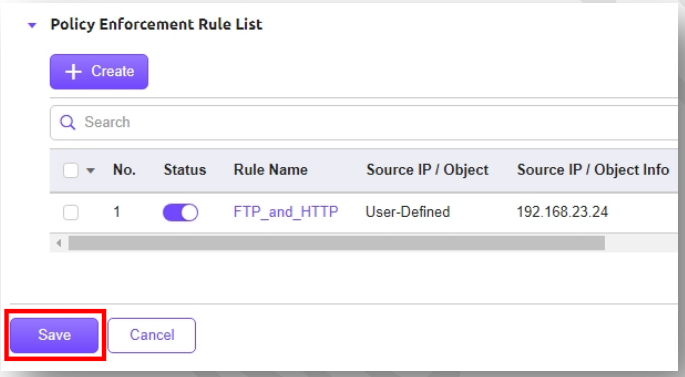


6. **Enable** the IPS Profile in Advanced Filter, choose the "**Default_IPS_Profile**" and Click "**OK**"

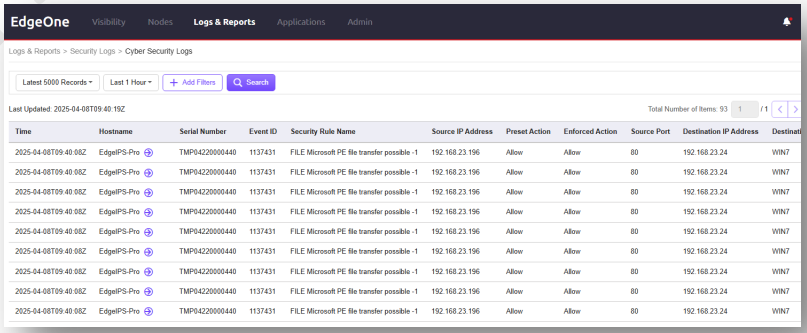
Items	Descriptions

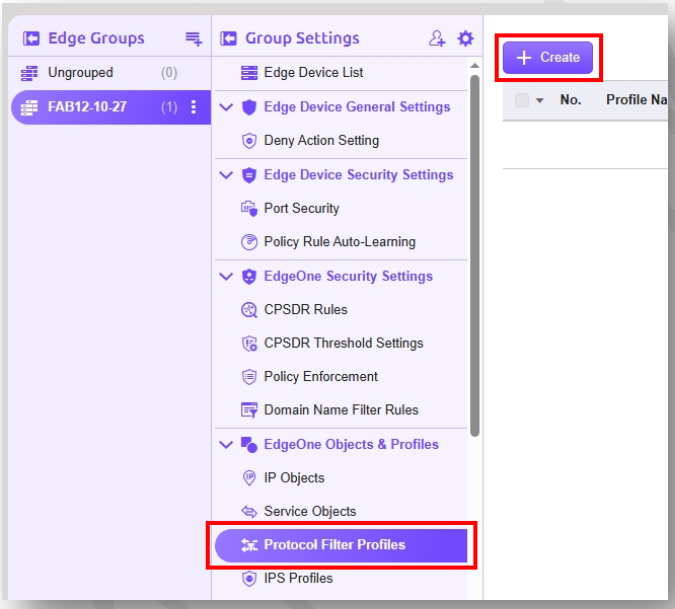
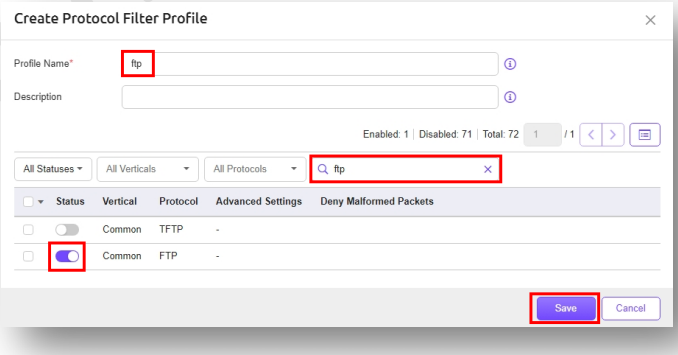


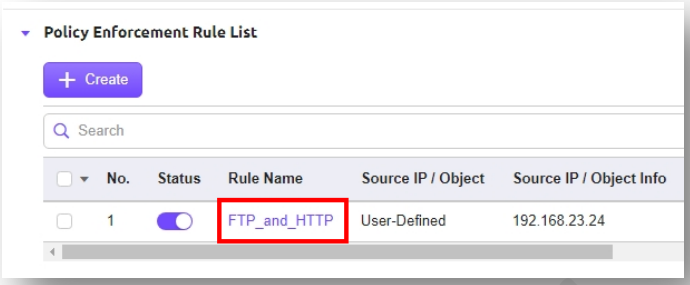
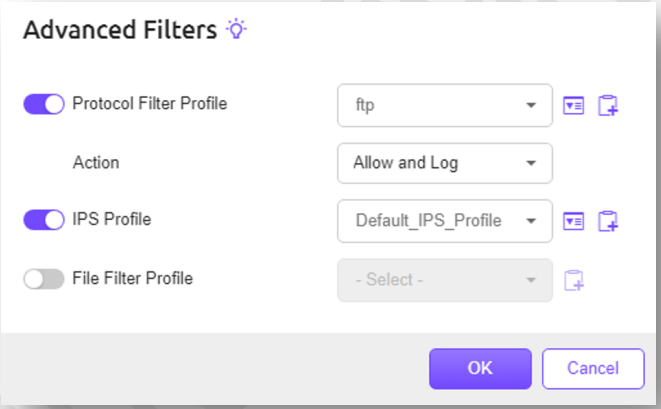
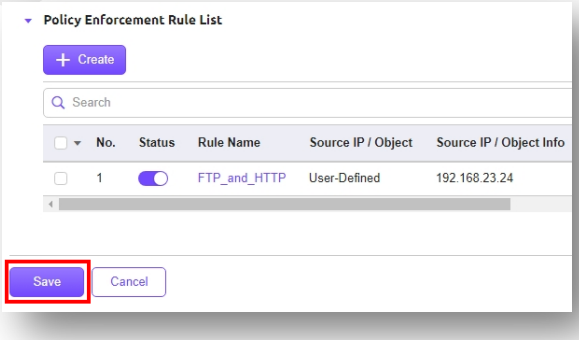
7. Click "**Save**" to save the policy.

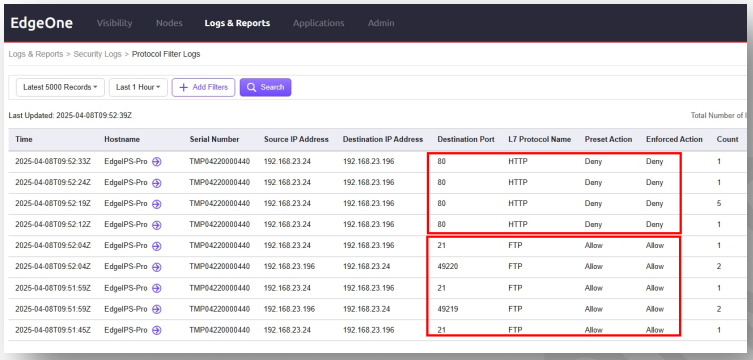
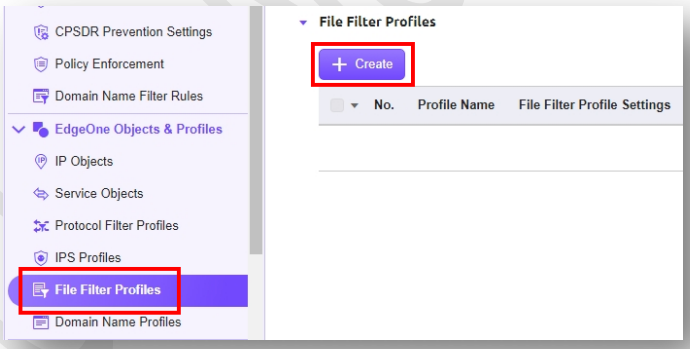


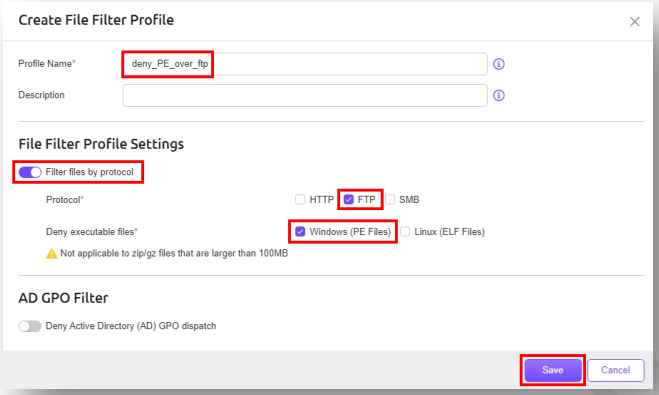
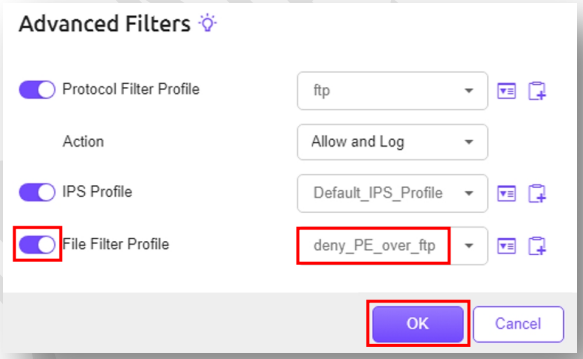
8. Execute the "**Download files over FTP**" SOP. The file download over FTP should be **successful**.
9. Execute the "**Download files over http**" SOP. The file download over FTP should be **successful**.
10. Those download activities will trigger the suspicious PE file transfer cyber security events.
11. Check the **Logs & Reports > Security Logs > Cyber Security Logs**



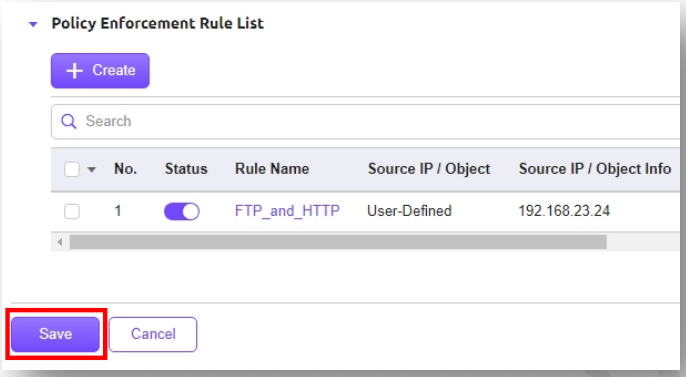
Items	Descriptions
	Configure Advanced Filter – Protocol filter Profile to allow ftp protocol only 1. Go to **Nodes > EdgeIPS Family > EdgeIPS Pro** 2. Select the Edge Group "**FAB12-10-27**". 3. Choose the **Policy Enforcement** 4. Choose the **Protocol Filter Profiles** 5. Click on the "**+ Create**" button  6. Profile Name: **ftp.** 7. Search: **ftp**, **Enable** FTP protocol, and click the "**Save**" button.  8. Add the Protocol Filter Profile to the Policy Enforcement. Choose the **Policy Enforcement** 9. Select the rule set: "**Port_Pair-01**"

Items	Descriptions
	10. Select the Rule: “**FTP_and_HTTP**”.  11. Enable the **Protocol Filter Profile**. 12. Choose “**ftp**” profile. 13. Set Action to “**Accept and Log**” 14. Click “**OK**”  15. Click “**Save**” to enforce the policy.  16. Execute the “**Download files over FTP**” SOP. The file download over FTP should **successful**.

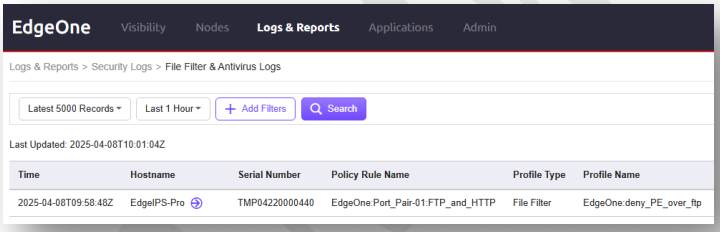
Items	Descriptions
	17. Execute the "**Download files over http SOP**". The file download over HTTP should fail because only FTP is allowed to pass through. 18. Check the **Logs & Reports > Security Logs > Protocol Filter Logs**.  **Configure Advanced Filter – File Filter Profile to deny PE file transfer over ftp** 1. Go to **Nodes > EdgeIPS Family > EdgeIPS Pro** 2. Select the Edge Group "**FAB12-10-27**". 3. Choose the "**File Filter Profiles**" and click on the "**+ Create**" button.  4. Profile Name: "**deny_PE_over_ftp**. 5. Enable the "**File Filter by Protocol**". 6. Check the Protocol: **FTP.** **7.** Deny Executable File(s): **Windows (PE File).**

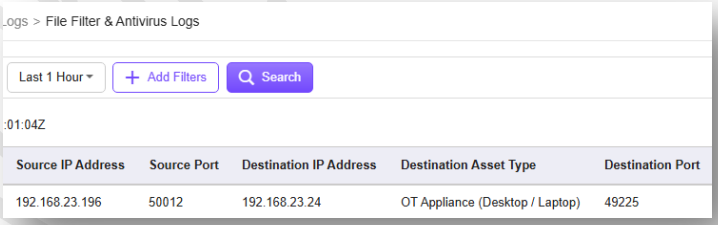
Items	Descriptions
	 8. Add the **File Filter Profile** to the Policy Enforcement. Choose the **Policy Enforcement** 9. Select the ruleset: **Port_Pair-01** 10. Select the rule: **FTP_and_HTTP** 11. Enable the File Filter Profile and choose the "**deny_PE_over_ftp**" profile. 12. Click "**OK**"  13. Click "**Save**" to enforce the policy.

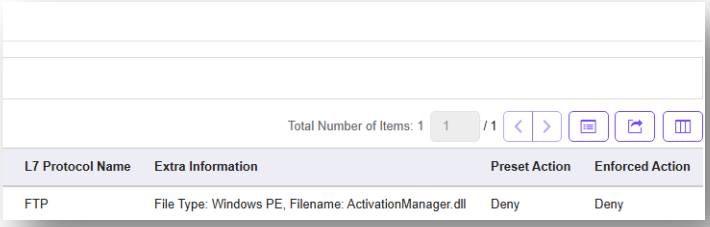
Items	Descriptions



14. Execute the "**Download files over FTP SOP**". Only the download of the text file over FTP should be **successful**. The download of the PE file will fail with a "**Connection aborted**" message.
15. Go to **Logs & Reports > Security Logs > File Filter & Antivirus Logs**.







Items	Descriptions
	Also check the following logs. Could you explain the result of logs? 16. Policy Enforcement Logs 17. Cyber Security logs 18. Protocol Filter Logs
Result and Comments	EdgeIPS Pro 216 “Basic” and “Advanced” filter of Policy Enforcement Rule tested

TXOne Networks

TXOne Networks